

OpenLane2 Timing & Area Report

Design: crossbar | PDK: sky130A HD | Run: 2026-05-16

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(a) Clock Period & Worst-Case Slack

Parameter	Value
Clock period (configured)	10.00 ns
Worst-case corner	max_ss_100C_1v60 (slow silicon, 100 °C, 1.60 V)
Setup slack — worst corner	−0.613 ns (VIOLATED)
Setup slack — nominal (TT 25 °C 1.80 V)	+2.469 ns
Setup slack — best (FF −40 °C 1.95 V)	+3.689 ns
Total negative slack (TNS)	−1.695 ns across 16 endpoint violations
Hold slack (all corners)	+4.09 ns — no violations

Note: The design is purely combinational (0 flip-flops). OpenROAD created a synthetic `__VIRTUAL_CLK__` to bound the I/O paths. The `−0.613 ns` violation means the longest combinational cone cannot complete within the 7.75 ns effective budget (10 ns − 2 ns output delay − 0.25 ns uncertainty) at the worst PVT corner. The design passes comfortably at nominal and fast corners.

(b) Critical Path

Source → Sink: v0[2] (input port) → i3[9] (output port i3, bit 9)

No source/sink *registers* exist — this is a pure combinational cone from primary input v0[2] to primary output i3[9]. The tool treats I/O ports as virtual register endpoints under `__VIRTUAL_CLK__`.

Path Trace (max_ss_100C_1v60, 8.363 ns arrival)

Stage	Cell	Type	Delay (ns)	Cumulative (ns)
Input buffer	input3	clkbuf_4	0.296	2.305
OR-4 (dominant)	_525_	or4_4	1.122	3.429

Stage	Cell	Type	Delay (ns)	Cumulative (ns)
OR3-AND-1	_575_	o31a_1	0.700	4.133
OR-3b	_820_	or3b_4	0.817	4.951
AND21-OI	_836_	a21boi_2	0.358	5.310
XOR-2	_839_	xor2_1	0.350	5.660
XOR-2	_841_	xor2_1	0.444	6.104
OR21-AND	_842_	o21a_1	0.404	6.508
AND-2b	_861_	and2b_1	0.437	6.946
OR41-AND	_864_	o41a_1	0.767	7.713
OR21-AND	_865_	o21a_1	0.393	8.106
Output buffer	output72	buf_6	0.254	8.361
Total arrival				8.363 ns
Required time				7.750 ns
Slack				−0.613 ns

Dominant Cell Types by Delay Contribution

Rank	Cell Type	Path Delay	Notes
1	or4_4	1.122 ns	High fanout ($\times 7$), large cap load, slow OR at SS corner
2	or3b_4	0.817 ns	Drives 3 fanout at 100 °C / 1.60 V
3	o41a_1	0.767 ns	Complex 4-input OR-AND gate

The 4-input OR (or4_4) contributes **13.4%** of total path delay. All three dominant stages are wide OR/AND-OR structures, consistent with the signed addition logic (**s0+s1−s2−s3** etc.) resolving carry/sign bits through deep OR trees.

(c) Cell Area — Top Contributors

Metric	Value
Total standard cell area (post-PnR)	4,200.28 μm^2
Total instances (logic)	764
Core utilisation	34.8% of 12,061.6 μm^2
Sequential elements	0 (100% combinational)
Die bounding box	122.24 μm $\times$ 132.96 μm

Top 3 by Instance Count (Logic Cells from Synthesis)

Rank	Cell	Count	Function
1	sky130_fd_sc_hd__xnor2_2	68	2-input XNOR — sign/bit comparison in adder network
2	sky130_fd_sc_hd__nand2_2	54	2-input NAND — complement logic
3	sky130_fd_sc_hd__nor2_2	52	2-input NOR — complement logic

Cell Class Breakdown (Post-PnR)

Class	Count	% of logic instances
Multi-input combinational	496	64.9%
Timing repair buffers (added by OpenROAD)	80	10.5%
Inverters	20	2.6%

Summary & Recommendation

The -0.613 ns violation is a **timing budget artefact** of the 2 ns output-delay assumption OpenROAD applies without an explicit SDC file. Adding a `SIGNOFF_SDC_FILE` with realistic I/O constraints, or reducing `CLOCK_PERIOD` to ~ 7 ns, would give an accurate closure picture. At TT/nominal the design carries $+2.47$ ns of positive slack — it is functionally fast.

If timing closure at the SS corner is required, the `or4_4` $\rightarrow$ `or3b_4` $\rightarrow$ `o41a_1` chain is the primary target: restructure the adder tree (e.g. carry-select or prefix-tree adder) or replace the deep OR fan-in with balanced two-level logic.

Check	Result
DRC	PASS
LVS	PASS
Antenna	PASS
Setup (TT nominal)	PASS
Setup (SS 100 °C worst)	FAIL (-0.613 ns, SDC artefact)
Hold (all corners)	PASS